ET Gen AI Hackathon 2026 — PS 6

Market Nerve

Autonomous Signal Intelligence for Indian Equities

Problem

14 crore Indian demat account holders flying blind — reacting to WhatsApp tips, missing exchange filings, unable to interpret technicals, managing portfolios on gut feel. ET Markets has the data. Nobody has built the intelligence layer to turn it into money-making decisions.

Vision

A multi-agent autonomous system that watches Indian markets in real-time, surfaces signals before they become news, explains chart patterns in plain English with back-tested confidence, and auto-generates daily market intelligence — all without a human analyst.

Team	Track	Stack	Prize Target
Up to 4 members	AI for the Indian Investor	Multi-agent / LangGraph / Next.js	**₹5,00,000 — First Place**

1. The Problem

India's retail investor explosion is one of the defining economic stories of this decade. Demat accounts crossed 14 crore. SIP flows hit record highs every month. But the tooling available to the average Indian investor is stuck in 2010.

1.1 The Structural Gap

Three broken realities coexist in the Indian equity market today:

The Data Exists

NSE/BSE publish bulk deals, insider trades, SAST filings, AGM minutes, quarterly results, and management commentary in real time. Every signal a sophisticated fund manager reads is publicly available.

Nobody Reads It

A retail investor with a full-time job cannot monitor 2,000 NSE-listed companies, track 50+ data sources, back-test chart patterns, and cross-reference FII/DII flows simultaneously. The cognitive load is impossible.

News Is Too Late

By the time ET Markets publishes a story about an unusual bulk deal or insider buy, the price has already moved. The signal and the news cycle are separated by 12–48 hours. Retail always arrives second.

1.2 What Currently Exists

Every existing solution fails in a specific, measurable way:

Category	Current solutions	Why they fail
Screeners	Screener.in, Ticker.in, Trendlyne	Passive filters. No proactive alerting, no signal synthesis, no narrative generation.
Brokerages	Zerodha Kite, Groww, Angel One	Transaction-focused. Research is generic tips, not personalized signal intelligence.
AI Chatbots	ChatGPT, Perplexity, ET Market ChatGPT	Reactive and generic. No live NSE data, no portfolio awareness, no back-tested pattern intelligence.
PMS / Advisory	SEBI-registered advisors, PMS funds	₹25,000+ per year. Serves only HNIs. 95% of Indian investors are excluded by price alone.

The Gap in One Sentence

India has 14 crore investors and zero autonomous, proactive, back-tested, portfolio-aware signal intelligence systems available to any of them.

2. The Product — Market Nerve

Market Nerve is not a dashboard, a chatbot, or a screener. It is an autonomous financial intelligence engine — a multi-agent system that continuously watches Indian equity markets, detects patterns humans cannot see at scale, synthesizes signals from 50+ data sources, and delivers intelligence that is actionable, attributed, and back-tested.

2.1 Core Design Principles

Proactive, not reactive	The system surfaces intelligence before users ask. Alerts arrive before news. The machine watches so the investor doesn't have to.
Back-tested confidence	Every signal comes with its historical win rate on that specific stock. Not generic pattern confidence — this exact pattern, this exact ticker, since listing.
Portfolio-aware intelligence	Every insight is contextualized to what the user actually holds. 'This FII sell-off in HDFC Life affects your portfolio by 2.3%' is worth 10x more than a generic market alert.
Source-cited transparency	Every claim links to its primary source — NSE filing, SEBI disclosure, ET Markets article. Trust is earned by showing the evidence, not asserting the conclusion.
Vernacular-ready	All intelligence is available in Hindi, Tamil, Telugu, and Bengali. Not translated — culturally adapted. A wheat farmer in Punjab and a salaried engineer in Chennai receive the same signal calibrated to their context.

2.2 The Four Intelligence Modules

Market Nerve is built as four deeply integrated modules. Each module is independently valuable. Together, they create something that cannot be replicated by any single-model product.

Module 1 — Signal Scout (Opportunity Radar)

Signal Scout is the engine that turns India's public financial data firehose into a curated stream of pre-news intelligence. It runs continuously, ingesting every exchange filing, corporate disclosure, and regulatory announcement as it is published.

What it watches:

- NSE/BSE bulk deal and block deal records — cross-referenced with historical patterns to detect anomalies
- SEBI insider trading disclosures — promoter and institutional buy/sell clustering detection
- SAST (Substantial Acquisition of Shares) filings — stake building and creeping acquisitions
- AGM minutes and management commentary transcripts — NLP extraction of sentiment shifts
- Quarterly results — beat/miss detection against consensus, segmental revenue inflections
- FII/DII daily flow data — sector rotation signals and institutional positioning shifts
- Pledging data changes — promoter distress and deleveraging pattern detection
- Related party transactions, exceptional items, auditor changes — governance signal layer

What it produces:

Example Signal Output

SIGNAL DETECTED: Persistent Systems (PERSISTENT) — Promoter has acquired shares in 4 of the last 5 market dips over 18 months. Current -7.3% drawdown represents the 5th such opportunity. Historical outcome: stock recovered above acquisition price within 60 trading days in 4/4 prior instances. Source: NSE bulk deal filings, SEBI promoter disclosures. Confidence: HISTORICAL PATTERN MATCH (not a recommendation).

The critical innovation here is not detection — screeners detect. It is synthesis: the agent cross-references 8 data sources, identifies the historical pattern, computes the back-tested outcome, and writes a source-cited plain-English alert in under 3 seconds of autonomous reasoning.

Module 2 — Pattern Mind (Chart Intelligence Engine)

Pattern Mind performs real-time technical pattern recognition across the full NSE-listed universe, with one capability no existing product offers: stock-specific historical back-testing delivered in plain English.

Patterns detected in real-time:

- Breakout from consolidation zone
- Double bottom / double top
- Head and shoulders formations
- Fibonacci retracement levels
- RSI/MACD divergence signals
- Support/resistance tests with volume confirmation
- Bollinger Band squeeze breakouts
- Cup-and-handle formations
- Golden cross / death cross
- Volume spike anomalies
- Price-volume divergence
- Sector relative strength breaks

The Back-Test Advantage

Most chart tools say: 'HDFC Bank has formed a Golden Cross.' Market Nerve says: 'HDFC Bank has formed a Golden Cross. This specific pattern has appeared 9 times in HDFC Bank since

2010. In 7 of 9 cases, the stock was higher 30 trading days later by an average of 8.4%. The 2 failure cases both coincided with broad market corrections (>5% Nifty drawdown). Current market context: Nifty is in a consolidation range, no broad correction signal active.'

Module 3 — Portfolio Lens (Next-Generation Market Intelligence)

Portfolio Lens replaces the generic financial chatbot with something categorically different: a portfolio-aware conversational analyst that uses the user's actual holdings as the lens through which all market intelligence is filtered.

Onboarding and data ingestion:

- CAMS/KFintech statement upload → automatic portfolio reconstruction
- Zerodha/Groww/Angel One CSV export → direct equity holdings import
- Manual entry for any asset class
- Zero retention mode: portfolio data used only for session-scoped context, not stored

What it can answer:

Sample question	What makes this hard
'Which of my mutual funds overlap most with my direct equity holdings?'	Requires: portfolio parsing + fund holdings data + overlap matrix computation + ranked output
'How exposed am I to a Reliance Industries earnings miss?'	Requires: direct + indirect (MF) Reliance exposure calculation + earnings sensitivity + portfolio beta
'If RBI hikes rates by 50bps tomorrow, what happens to my portfolio?'	Requires: rate sensitivity by sector + duration of debt MFs + historical sector correlation to rate cycles
'My XIRR is 14.2% — is that good?'	Requires: benchmark comparison (Nifty 50, category average) + risk-adjusted return + time period context

Every answer is source-cited with primary ET Markets data. Multi-step reasoning chains are shown — not just the answer, but the logic that produced it.

Module 4 — Story Engine (Autonomous Market Media)

Story Engine is the most visible, most demonstrable, and most emotionally resonant module in Market Nerve. It auto-generates broadcast-quality market intelligence media — video, visual narratives, and interactive story arcs — with zero human involvement.

Daily Market Wrap (video):

Every weekday at 5:15 PM, Story Engine ingests the day's market data and generates a 60–90 second animated video: opening/closing Nifty movement, top gainers and losers with price

action, FII/DII flows with race-chart visualization, sector rotation summary, and one AI-identified headline signal. The video renders in under 90 seconds. Format: shareable MP4 + embedded web player. No editor, no voiceover artist, no production team.

Story Arc Tracker:

Users select any ongoing business story — Adani Group, Byju's debt saga, HDFC merger integration, Zomato profitability journey. The AI constructs a complete interactive visual narrative: chronological timeline with source-cited events, sentiment shift tracking across ET coverage, key player network map, contrarian perspectives surfaced from analyst reports, and a 'what to watch next' predictive summary.

IPO Intelligence Dashboard:

Live GMP tracking, subscription status, allotment probability calculator, and AI-written plain-English breakdown of every active IPO's financials — formatted for an investor with no accounting background.

3. System Architecture

3.1 High-Level Overview

Market Nerve is built as a distributed multi-agent system. Four specialized agents run asynchronously, each owning a specific data domain. A central Orchestrator manages coordination, routing, and the auditable decision trail. All agent outputs flow through a shared Event Bus, enabling cross-agent signal enrichment.

Agent	Responsibility	Primary inputs	Primary outputs	Reasoning model
Signal Scout	*Market signal detection and synthesis*	NSE/BSE feeds, SEBI filings, FII/DII data, AGM transcripts	Ranked signal alerts with back-test metadata	Claude Sonnet + LangGraph tool calls
Pattern Mind	*Technical analysis across full NSE universe*	OHLCV data via yfinance/NSE Python, historical price data	Pattern matches with stock-specific win rate	TA-Lib + Claude for narrative generation
Portfolio Lens	*Portfolio-contextualized Q&A and analysis*	User CAMS/broker export, live market data, MF holdings DB	Source-cited multi-step analysis responses	Claude Sonnet with extended context
Story Engine	*Autonomous financial media generation*	All agent outputs, ET Markets data feed, IPO filings	Video files, story arc HTML, IPO dashboards	Claude + Remotion.js + D3.js
Orchestrator	*Agent coordination and audit logging*	All agent events, user requests, system health	Task routing, conflict resolution, audit trail	LangGraph supervisor pattern

3.2 Technology Stack

Layer	Technology	Rationale

Frontend	Next.js 14 (App Router) + TypeScript + TailwindCSS + D3.js	SSR for SEO, React server components for data-heavy pages, D3 for custom chart rendering
Agent Framework	LangGraph (Python) + Anthropic Tool Use API	LangGraph enables stateful multi-agent graphs with explicit edges; Tool Use enables structured data extraction from unstructured filings
Core LLM	Claude Sonnet 4 via Anthropic API	Best reasoning-per-cost for multi-step financial analysis; extended context handles full CAMS statements + portfolio + live data
Technical Analysis	TA-Lib (Python) + pandas-ta	Battle-tested indicator library; 150+ indicators; used by professional quant desks
Market Data	NSE Python unofficial API + yfinance + Screener.in scraper	NSE Python for real-time; yfinance for historical OHLCV; Screener for fundamental data
Exchange Filings	NSE/BSE direct filing parsers (Python) + SEBI EDGAR scraper	Structured parsing of bulk deals, SAST, insider trade disclosures at source
Video Generation	Remotion.js + React + FFmpeg	Code-first video generation; every frame is a React component; programmatic control of every visual element
Job Queue	BullMQ + Redis	Reliable queue for async agent tasks; dead letter queues for failed jobs; rate limiting for API calls
Backend API	FastAPI (Python) + Hono (Node.js)	FastAPI for Python agent services; Hono for edge-deployed lightweight endpoints
Database	PostgreSQL (primary) + Redis (cache/pub-sub) + Pinecone (vector)	PostgreSQL for structured financial data; Redis for live feed pub-sub; Pinecone for semantic search over filings
Auth + Portfolio	Clerk.dev + AES-256 encrypted portfolio storage	Zero-knowledge option: portfolio data can be session-only, never persisted
Deployment	Vercel (frontend) + Railway (Python agents) + Cloudflare Workers (edge)	Frontend on edge CDN; agents on always-on containers; critical latency paths on Workers
Monitoring	Langfuse (LLM observability) + Sentry + Grafana	Langfuse traces every agent reasoning step; Grafana for system health; Sentry for error tracking

3.3 Data Flow Architecture

Ingestion Layer

Five concurrent data pipelines feed the system:

- Real-time exchange feed: NSE market data API polling every 60 seconds during market hours (9:15 AM – 3:30 PM IST)
- Filing parser: BSE corporate filings and NSE announcements scraped and classified within 90 seconds of publication
- FII/DII flow aggregator: SEBI provisional FII/DII data parsed daily post 6 PM
- Fundamental data sync: Screener.in quarterly results, financial ratios updated post-results within 2 hours
- News sentiment layer: ET Markets article NLP for management commentary sentiment extraction

Processing Layer

All raw data flows into a Redis pub-sub channel where agents are subscribed to their relevant event types. Signal Scout listens to filing events. Pattern Mind listens to OHLCV update events. The Orchestrator listens to all channels and maintains the global state graph.

Audit Trail

Auditability Requirement

Every agent decision — every signal surfaced, every pattern flagged, every portfolio analysis produced — is logged with: timestamp, input data snapshot, reasoning chain (the Claude tool use trace), output generated, and confidence metadata. This audit trail is queryable by the user. Judges and institutional users can inspect exactly why any alert was triggered.

4. API Design

4.1 Core Endpoints

Method	Endpoint	Description
GET	/api/signals	Returns ranked signal alerts for a given time window. Query params: limit, sector, min_confidence, since.
GET	/api/signals/{ticker}	All historical signals for a specific NSE ticker, with back-test metadata.
POST	/api/signals/subscribe	Subscribe to push alerts for a watchlist of tickers or sectors.
GET	/api/patterns/{ticker}	All detected technical patterns for a ticker with stock-specific historical win rates.
GET	/api/patterns/scan	Real-time pattern scan across full NSE universe. Filter by pattern type, sector, market cap band.
POST	/api/portfolio/analyze	Upload CAMS/broker CSV. Returns portfolio reconstruction: XIRR, overlap matrix, sector exposure, benchmark comparison.
POST	/api/portfolio/query	Natural language Q&A against an uploaded portfolio. Returns source-cited multi-step analysis.
GET	/api/story/video/latest	Returns the latest auto-generated daily market wrap video (MP4 + web player embed).
GET	/api/story/arc/{query}	Generate or retrieve a Story Arc for any company, sector, or market narrative.
GET	/api/ipo/active	All active IPOs with GMP, subscription status, allotment probability, AI-written breakdown.
GET	/api/audit/{signal_id}	Full audit trail for any signal: input data, reasoning chain, confidence metadata.
GET	/api/health	System health: agent status, data freshness, API latency percentiles.

4.2 WebSocket Streams

Stream	Payload
ws://market-nerve.io/live/signals	Real-time signal events as they are detected. Filter by watchlist client-side.
ws://market-nerve.io/live/patterns	Pattern detection events as they cross threshold during market hours.
ws://market-nerve.io/live/filings	Raw filing classification events: filing type, ticker, parsed metadata, URL.

5. User Experience Design

5.1 Information Architecture

/ (Home)	Live signal feed — ranked by freshness and confidence. Three-column layout: signal summary, source metadata, quick portfolio impact if holdings imported.
/signals	Signal Scout deep-dive. Filter by sector, signal type, confidence band. Every signal links to full audit trail.
/patterns	Pattern Mind interface. NSE universe grid with real-time pattern overlay. Click any stock → full pattern detail with back-test history rendered as inline chart.
/portfolio	Portfolio Lens. Upload flow → portfolio X-ray dashboard → conversational Q&A interface. Split-pane: analysis panel left, chat right.
/story	Story Engine. Today's video auto-plays. Below: Story Arc search. Type any company → AI builds interactive timeline.
/ipo	IPO Intelligence. Live GMP tracker, subscription meter, allotment probability, AI-written plain-English breakdown per IPO.
/audit/{id}	Full audit trail for any signal or analysis. Input data → reasoning steps → output. One-click share.
/vernacular	Language selection UI. All intelligence available in Hindi, Tamil, Telugu, Bengali. Toggle persists across session.

5.2 The Onboarding Flow

The single most important UX moment in Market Nerve is the first 90 seconds. The user must understand what they are looking at, feel the intelligence is real, and want to connect their portfolio. This is the design of that moment:

0s	Landing page loads. Live signal feed is visible immediately — no login required. Three real signals from today, already back-tested, already source-cited. The intelligence is not behind a paywall.

15s	A small banner: 'Upload your portfolio to see which of these signals affect your holdings.' One button. No account required.
30s	User uploads CAMS statement (drag-and-drop). Progress indicator while portfolio is parsed — 'Reconstructing your portfolio...'
60s	Portfolio X-ray completes. The live signal feed now shows portfolio impact badges in red/green next to each alert. Personalization is instant and visible.
90s	A modal appears with the portfolio's Money Health Score — one number, six sub-scores, three recommended actions. The user now has a reason to return tomorrow.

5.3 Design Language

- Typography: Space Grotesk for UI labels and numbers (financial readability); Georgia for editorial content and signal narratives. The contrast between the two typefaces signals the distinction between data and intelligence.
- Color system: Dark charcoal (#111111) base. Teal (#1D9E75) for confirmed signals and positive states. Amber (#B97B10) for patterns and uncertain states. Red (#991F1F) for warnings and negative signals. White space is generous — financial data already has high cognitive load.
- Motion: Signal arrivals animate in with a 200ms slide from right. Portfolio impact badges pulse once on load. Video generation shows a progress ring that fills in real-time. All motion is purposeful — it encodes information, not decoration.
- Density: Three information tiers. Default view: signal headline + confidence + age. Expanded: full back-test metadata + source list. Audit view: complete reasoning trace. Users choose their depth.

6. Agent Logic & Reasoning Design

6.1 Signal Scout Agent — Reasoning Chain

Every signal produced by Signal Scout must pass through a five-stage reasoning chain before it reaches the user. This is enforced by the LangGraph graph structure — stages cannot be skipped.

#	Stage	Logic
1	**Event detection**	Raw filing or data point arrives via pub-sub. Agent classifies event type (bulk deal, insider trade, AGM, etc.) and extracts structured metadata using Claude tool call.
2	**Anomaly scoring**	Extracted metadata compared against 24-month historical baseline for that ticker. Z-score computed. Events below 1.5 standard deviations are discarded. Only genuine anomalies proceed.
3	**Cross-source enrichment**	Anomalous event is enriched: What else happened in this ticker in the last 7 days? Any related sector events? Any management commentary shifts? All enrichments are sourced and logged.
4	**Historical back-test**	Pattern matching against full available history for that ticker. Win rate, average return, time to recovery, market context during failures — all computed and attached.
5	**Plain-English synthesis**	Claude writes the signal narrative: what happened, why it's unusual, what history says, what to watch, with every claim linked to source. Output undergoes a self-consistency check — the agent reads back its own output and flags any unsupported claim.

6.2 Portfolio Lens — Context Window Management

The most technically demanding challenge in Portfolio Lens is fitting a complete portfolio into the Claude context window while leaving enough space for multi-step reasoning. The solution is a three-tier context compression strategy:

- Tier 1 — Structural summary: Holdings count, total value, asset class breakdown, XIRR. Always in context. 500 tokens.
- Tier 2 — Sector exposure matrix: Percentage allocations by sector for direct equity and each MF separately. In context for sector-related queries. 800 tokens.

- Tier 3 — Individual holding details: Full holding-by-holding breakdown. Retrieved via tool call on demand — only the relevant holdings are fetched for each query. Prevents context bloat.

Self-Verification Rule

Every Portfolio Lens response that includes a number (percentage, XIRR, rupee value, count) must be verified by a second pass where the agent asks itself: 'Is this number derivable from the user's actual portfolio data?' If no, the response is rejected and regenerated. This eliminates hallucinated financial figures — the single most dangerous failure mode in a financial AI product.

6.3 Story Engine — Video Generation Pipeline

Step	Implementation
Data snapshot (T+0s)	At 5:15 PM, BullMQ job triggers. System takes a snapshot of: Nifty/Sensex close, top 5 gainers/losers, FII/DII provisional data, sector heatmap, top signal from Signal Scout.
Script generation (T+5s)	Claude generates the 90-second narration script from the data snapshot. Voice-over text, scene descriptions, data to visualize per scene.
React component rendering (T+20s)	Remotion.js renders each scene as a React component tree. D3.js powers the race charts and heatmaps. All animations are programmatic.
Video compilation (T+60s)	FFmpeg compiles rendered frames into H.264 MP4. Resolution: 1080x1920 (vertical for mobile) + 1920x1080 (horizontal for web). Both generated in parallel.
Distribution (T+90s)	Video uploaded to CDN. WebSocket broadcast to all connected clients. Push notification sent to subscribed users. ET Markets embed code generated.

7. Build Plan & Team Structure

7.1 Recommended Team Composition

Role	Primary ownership	Scope
Lead / AI Architect	Agent system + LLM integration	LangGraph graph design, Claude tool use chains, agent prompt engineering, signal reasoning logic, audit trail system
Backend Engineer	Data pipeline + APIs	NSE/BSE parsers, SEBI filing scrapers, PostgreSQL schema, Redis pub-sub, FastAPI endpoints, BullMQ job queues
Frontend Engineer	Next.js UI + D3 visualizations	Signal feed UI, portfolio X-ray dashboard, pattern chart renderer, IPO tracker, WebSocket integration, vernacular toggle
Media / ML Engineer	Story Engine + Pattern Mind	Remotion.js video pipeline, FFmpeg integration, TA-Lib pattern detection, back-test computation engine

7.2 Two-Week Sprint Plan

Phase	Days	Deliverables
Foundation	**1–3**	NSE/BSE data pipeline live. PostgreSQL schema. Redis pub-sub. Signal Scout agent skeleton. LangGraph orchestrator running. Basic Next.js shell deployed.
Core Intelligence	**4–7**	Signal Scout fully functional — 8 data sources ingested, anomaly detection running, back-test engine live. Portfolio Lens CAMS parser + Claude Q&A end-to-end. Pattern Mind: 12 patterns detected in real-time across Nifty 500.
Media & UX	**8–11**	Story Engine: daily video pipeline live and generating test videos. Story Arc Tracker: 5 pre-built arcs (Adani, Zomato, HDFC Merger, Paytm, Jio Financial). Front-end signal feed, portfolio dashboard, pattern chart renderer all connected to live data.

Polish & Demo	**12–14**	IPO intelligence module. Vernacular layer (Hindi + Tamil minimum). Audit trail UI. Demo script rehearsed. Architecture document. 3-minute pitch video recorded. GitHub README. Impact model finalized.

7.3 Demo Priority Stack

If the build is behind schedule entering the final 3 days, deliver in this exact priority order:

1. Signal Scout live with 5 real signals from today. Source-cited. Back-tested. This is the heart of the product.
2. Portfolio Lens: CAMS upload → portfolio X-ray → 3 scripted Q&A flows that demonstrate multi-step reasoning.
3. Story Engine: One pre-generated market wrap video. Even if video generation is not live, one pre-rendered video demonstrates the concept and creates the WOW moment.
4. Pattern Mind: Beautiful UI with 5 pre-loaded pattern examples and historical back-test charts. Real-time detection is a bonus.
5. Vernacular layer, IPO module, Story Arc Tracker — polish features. Build only after 1–4 are solid.

8. Impact Model

8.1 Market Sizing

Total demat account holders (India, 2026)	**14 crore (140 million)**
Active equity investors (transacted in last 12 months)	**~5 crore (35% of total)**
Addressable segment (smartphone, ET Markets user)	**~2 crore**
Conservative penetration at Year 1	**0.5% = 1 lakh users**
Freemium conversion rate (industry benchmark)	**5% = 5,000 paid users**
Average revenue per paid user (₹99/month, ET Prime bundle)	**₹1,188/year**
Year 1 ARR at conservative penetration	**₹5.94 crore (~$700K)**
Year 3 at 2% penetration, ₹499/month premium tier	**₹84 crore (~$10M)**

8.2 Strategic Value to ET

Beyond subscription revenue, Market Nerve generates compounding strategic value for ET specifically:

- **Data moat reinforcement:** ET Markets' existing data becomes the competitive advantage. Every signal Market Nerve surfaces cites ET data as source — users learn to trust ET as the primary intelligence layer, not just a news publisher.
- **DAU driver:** Daily market wrap videos + morning signal digests are a pull mechanism that brings users to ET properties every single market day. ET's current engagement is event-driven; Market Nerve makes it habitual.
- **Premium upgrade path:** Free tier → Signal Scout only. ET Prime tier → all four modules. The intelligence hierarchy directly maps to ET's existing subscription funnel.
- **Vernacular reach:** The vernacular layer opens Market Nerve to tier-2/3 India — a segment ET Markets does not currently serve effectively. Hindi and Tamil alone add an addressable population of 50+ crore.
- **Avataar partnership fit:** Avataar's core thesis is agentic AI for enterprise transformation. Market Nerve is exactly that — autonomous agents driving measurable operational value (intelligence delivery at scale) for ET as an enterprise.

8.3 Cost Structure

LLM API (Claude Sonnet, ~50K tokens/signal × 500 signals/day)	**~₹15,000/month at scale**
Infrastructure (Vercel + Railway + Cloudflare)	**~₹8,000/month**
Market data (NSE Python + yfinance — open source)	**₹0**
Video rendering (Remotion + FFmpeg — self-hosted)	**~₹3,000/month server cost**
Total operational cost at 10,000 daily active users	**~₹26,000/month**
Revenue at 500 paid users (₹99/month)	**₹49,500/month**
Break-even point	**535 paid users**

9. Risks & Mitigations

Risk	Severity	Mitigation
NSE/BSE informal API rate limits	**Medium**	Primary: NSE Python library with respectful polling. Fallback: yfinance + Screener.in for historical. NSE XML official data feed for production.
LLM hallucination in financial context	**High**	Self-verification pass on all numbers. Explicit tool use for data retrieval — no LLM generates numbers from parametric memory. Every claim must have a source link or it is rejected.
SEBI regulatory compliance	**Medium**	All outputs carry the mandatory disclaimer: 'This is not SEBI-registered investment advice.' Signal outputs explicitly labeled as 'Historical pattern data, not a recommendation.'
Portfolio data privacy	**High**	Zero-retention mode as default — portfolio data lives only in session memory. Explicit user consent required for any persistence. AES-256 for anything stored.
Market data accuracy at demo time	**Low**	Three fallback data snapshots pre-loaded for each demo scenario. If live feed fails, switch to most recent snapshot. Failure mode is graceful, not catastrophic.
Video generation latency (>90s)	**Low**	Pre-generate demo videos for finale. Live generation is shown conceptually. One pre-rendered video is sufficient to prove the concept to jury.

10. Submission Checklist

10.1 Required Deliverables

GitHub Repository

Public repository. Commit history must show genuine build progression — not a single dump commit. README must include: system architecture overview, local setup instructions (one-command Docker Compose preferred), environment variables guide, and a link to the live demo. Every agent's LangGraph graph must be documented in code comments.

3-Minute Pitch Video

Open with the hook: three signals surfaced today that became ET news 48 hours later. Show Signal Scout live. Cut to Portfolio Lens Q&A. Cut to Story Engine generating a video in real-time. Close with the impact number: 14 crore investors. Zero products that do this. Market Nerve is the first. No slide decks — product only. Record at the end of the build, not the beginning.

Architecture Document

1–2 pages. Diagram showing: five agents (Signal Scout, Pattern Mind, Portfolio Lens, Story Engine, Orchestrator), the shared Event Bus, the data sources flowing in, the output channels flowing out. Second page: brief description of each agent's reasoning model, error handling approach, and audit trail mechanism. This document is evaluated by Avataar — they look for agent depth and autonomy design.

Impact Model

One page. Show the math: 14 crore demat holders → addressable segment → conversion assumptions → ARR at Year 1 and Year 3. Show the cost structure: LLM API cost per signal, infra cost, break-even user count. State all assumptions explicitly. The jury knows the assumptions are estimates — what they are evaluating is whether the logic holds and whether you understand your own unit economics.

10.2 Judging Criteria Mapping

Judging criterion	How Market Nerve scores

Innovation & Creativity	Back-tested, stock-specific pattern confidence is novel globally. Autonomous daily market video generation has not been productized for Indian retail at this scale. Story Arc Tracker is a new format.
Technical Implementation	Five-agent LangGraph system with shared event bus. Claude tool use for structured financial data extraction. Remotion.js + FFmpeg video pipeline. Full audit trail. Not a chatbot wrapper.
Feasibility & Scalability	Cost structure demonstrated to break even at 535 paid users. Architecture scales horizontally — agent workers are stateless. Redis pub-sub handles 10,000+ concurrent signal subscriptions.
Relevance to Problem Statement	Directly addresses every sub-problem listed in PS 6: Opportunity Radar ✓ , Chart Pattern Intelligence ✓ , Market ChatGPT Next Gen ✓ , AI Market Video Engine ✓ . Not a partial answer.
Documentation & Presentation	This document serves as the architecture reference. Pitch video: live product demo only. Impact model: explicit assumptions and unit economics. GitHub: one-command setup.

11. Finale Strategy — The 20-Minute Jury Room

The finale gives each team 15–20 minutes. The jury are ET veterans, market professionals, and Avataar's enterprise AI team. They have seen dozens of demos before yours. The only question that matters in those 20 minutes is: does this feel real?

11.1 The Minute-by-Minute Script

0:00–1:00	No slides. Open the product directly. Show the live signal feed. Say: 'These three signals appeared on Market Nerve yesterday. This one became an ET Markets story this morning. The machine knew first.' Let the silence land.
1:00–5:00	Signal Scout live demo. Pick a Nifty 500 stock the jury suggests — or have one pre-selected with a strong recent signal. Walk through the signal: what was detected, what data was used, what history says, what the source link is. Click the audit trail link. Show the reasoning chain.
5:00–9:00	Portfolio Lens. Upload the pre-prepared CAMS statement. Show the portfolio X-ray complete in under 60 seconds. Ask the question: 'Which of my mutual funds overlap most with my direct equity?' Show the multi-step answer with source citations. Ask a second question using a jury member's suggested stock.
9:00–13:00	Story Engine. Say: 'Every evening at 5:15 PM, this video generates automatically. No editor, no voiceover artist, no production team. It rendered in 80 seconds.' Play the video. Give the jury the QR code to the web player. Then show one Story Arc — Zomato or HDFC Merger. Let them scroll the timeline.
13:00–16:00	Impact. Show the number: 14 crore. Show the gap: zero products doing this. Show the unit economics: break-even at 535 paid users. Show the ET strategic fit: 'ET has the data. ET has the distribution. Market Nerve is the intelligence layer that was missing.'
16:00–20:00	Questions. Have the GitHub open. Have the audit trail visible for any signal they want to interrogate. Know the system well enough to modify a live parameter if asked. The best answer to 'can you show us X?' is doing it, not explaining why it would work.

11.2 The One Non-Negotiable

The Product Must Be Live

Do not present Market Nerve as a prototype, a mockup, or a concept. It must be running on a live URL with real NSE data flowing through it on the day of the finale. Judges who work in markets will ask for a real stock. They will look for a recent real signal. If the product is not live with real data, no amount of slide design or pitch skill will compensate. Invest the last 72 hours before the finale in hardening the live demo, not adding features.

Market Nerve

ET Gen AI Hackathon 2026 — Problem Statement 6

Build the intelligence layer Indian equity investors deserve.